

Pricing October

How the ticket market prices uncertainty during the 2026 MLB postseason: games that may never be played, series that can end in two nights, and a Yankees–Red Sox Wild Card that sells out before anyone knows who's pitching.

Michael Schwartz · Auburn University · BSBA Business Analytics, Computer Science minor · Class of 2028

12

TEAMS TRACKED

~45

POSSIBLE POSTSEASON GAMES

24/day

SNAPSHOTS PER GAME, EVERY HOUR

THE QUESTION

A regular-season ticket has weeks on sale and a known opponent. A postseason ticket can go on sale before the game exists ("Game 5, if necessary"), change opponent overnight, and vanish when a series ends early. **How does the market price that uncertainty, and where does that leave money on the table for the club?**

WHY IT MATTERS TO A TEAM

Postseason inventory is the highest-margin ticketing a club sells all year, and the one it has the least time to price. Every Strategy & Analytics group in the league watches the resale market in October. This project does what they do, on public data, and writes it up the way they would.

WHAT GETS BUILT

- An hourly data pipeline (Python → GitHub Actions → CSV → SQLite) that runs unattended for five weeks
- A SQL analysis layer: pacing, price curves, primary-vs-resale gap, series-state effects
- A regression that puts a number on the "if necessary" discount and the series-state premium
- A Tableau Public dashboard (4 views) and a one-page memo written for a VP of Ticketing
- A public GitHub repo with a README a hiring manager can read in three minutes

SKILLS ON DISPLAY

SQL · JOINS · CTES · WINDOW FUNCTIONS

PYTHON · REQUESTS · PANDAS · STATSMODELS

DATA ENGINEERING · APIS · ETL · SCHEDULING

CS · GIT · CI/CD · ERROR HANDLING · REPRODUCIBILITY

STATISTICS · REGRESSION · EVENT STUDY · ELASTICITY

TABLEAU PUBLIC DASHBOARD

TICKETING · DYNAMIC PRICING · PRIMARY VS SECONDARY · INVENTORY

EXECUTIVE MEMO · LINKEDIN POST · OUTREACH

TIMELINE

Sep 27–29

Keys, repo, first snapshot. Live before first pitch (Tue 2 pm ET).

Sep 29–Oct 1

Wild Card. Watch the pipeline, fix what breaks, fill the bracket.

Oct 3–10

Division Series. First SQL queries over Fall Break (Oct 8–9).

Oct 11–20

LCS. Regression model. First Tableau views (Oct 16–17).

Oct 23–31

World Series. Memo draft (Oct 24). README (Oct 30).

Nov 1–5

Final data. Publish Nov 2. Project-led outreach Nov 3–5.

WHO IT'S WRITTEN FOR

Primary: Mets Ticketing Strategy & Analytics (Jeff Manahan), Yankees Business Strategy summer program, MLB League Office Ticketing internship (deadline ~Dec 19). Secondary: every Tier-A team on the target list, because every one of them prices a playoff run the same way.

Status right now: scraper written and compiling (~Desktop/Pricing–October). Not yet running. It needs two free API keys and a GitHub repo, and only you can create those accounts. Page 2 is the checklist. First pitch is **Tuesday, Sep 29, 2:00 pm ET** (Phillies at Braves). Every hour before that is a data point we keep.

WHAT'S IN THIS BLUEPRINT

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What I need from you, in order

Everything below is something I can't do for you: it creates an account in your name. Each one is about five minutes. Do them in order; #5 is the one that turns the lights on.

BEFORE FIRST PITCH · DEADLINE TUE SEP 29, 1:00 PM ET · IDEALLY MONDAY NIGHT

	Item	Exactly what to do	What I do with it
☐	1. SeatGeek client ID resale-market data	Go to seatgeek.com/account/develop (log in or make a free SeatGeek account). Create an app: name it <i>pricing-october</i> , any URL. Copy the Client ID . If the page won't create apps anymore, tell me: we fall back to Ticketmaster only and the project still works.	Every hour: lowest / average / median / highest resale price and listing count for every postseason game.
☐	2. Ticketmaster consumer key primary-market data	Go to developer.ticketmaster.com → Get your API key → create an account → create an app. Copy the Consumer Key . (The Discovery API is free; 5,000 calls/day is far more than we need.)	Every hour: the club's own price range and on-sale status, so we can measure the primary-vs-resale gap.
☐	3. Fill the .env file	In ~/Desktop/Pricing-October, duplicate <code>.env.example</code> as <code>.env</code> and paste the two keys after the equals signs. Don't paste keys into chat . The file is git-ignored and never leaves your Mac.	I run the first snapshot from your machine and fix whatever the APIs complain about.
☐	4. GitHub repo + secrets	github.com → New repository → name pricing-october , Public , no README (we have one). Then Settings → Secrets and variables → Actions → New repository secret, twice: name SEATGEEK_CLIENT_ID, then TM_API_KEY, each with its key as the value.	The hourly job runs on GitHub's servers, not your laptop. Secrets keep the keys out of the code.
☐	5. Tell me "keys are in"	That's it. I initialize git, run the first snapshot, fix errors, and hand you one <code>git push</code> command. Then we watch the Actions tab for the first green check.	Pipeline live. Bracket page and schedule page take over from here.

DURING THE POSTSEASON · 2 MINUTES A DAY, PLUS THE BLOCKS ALREADY IN YOUR GAME PLAN

	When	What
☐	Every morning	Open the repo's Actions tab. Green check = data is flowing. Red X = screenshot it to me the same day. A missed day is a hole in the price curve we can't fill later.
☐	After every series	Fill in the bracket (p. 4) and mark the schedule (p. 5). Note anything weird you saw as a fan: a rainout, a pitcher announcement, a viral sellout. Those become the annotations on the dashboard.
☐	Thu Oct 8 · Fall Break	3-hour block: run <code>build_db.py</code> , write the first five queries (p. 9). This is the SQL you've been learning, on your own data.
☐	Fri Oct 16 – Sat Oct 17	Create a free Tableau Public account (public.tableau.com). Build the first two dashboard views (p. 10).
☐	Sat Oct 24	Memo draft (p. 11). Send it to me; I'll be the VP of Ticketing who tears it apart.
☐	Fri Oct 30 – Sat Oct 31	README with screenshots. Ask one Sports Analytics Club officer or professor to critique it.
☐	Mon Nov 2	Publish: LinkedIn post, dashboard link, one finding. Tag the Sports Analytics Club.
☐	Tue Nov 3 – Thu Nov 5	Project-led outreach (p. 11): Mets, Yankees, MLB, Braves, Rays, then the rest of Tier A.

ACCOUNTS YOU'LL END UP WITH (ALL FREE)

- SeatGeek developer app
- Ticketmaster developer app
- GitHub (public repo, Actions enabled by default)
- Tableau Public (mid-October)

WHAT YOU NEVER HAVE TO DO

- Leave your laptop on. GitHub runs the job.
- Write the scraper. Done. You'll read it, and we'll walk through it so you can explain every line in an interview.
- Learn Tableau from scratch. Four views, one data source. We build it together.

The system, end to end

Seven stages. Each one is a thing a recruiter can point at and ask you about. The tags say which skill it proves.

1

Two public APIs
SeatGeek /2/events (resale stats) and Ticketmaster Discovery v2 (primary priceRanges). Auth by query-string key.
APIS
PRIMARY VS RESALE

2

scrape.py
Pages through both APIs, filters to postseason titles by regex, flattens JSON to flat rows, appends to CSV. One source failing doesn't kill the other.
PYTHON
ERROR HANDLING

3

GitHub Actions
Cron at :07 every hour. Installs Python, runs the script with secrets injected, commits new rows. Zero servers, zero cost.
SCHEDULING
CI/CD · SECRETS

4

CSV in git
Append-only snapshot log. Every commit is a timestamped audit trail; the repo's history is the dataset's lineage.
VERSION CONTROL
DATA LINEAGE

5

SQLite
build_db.py loads both tables, indexes on (event, time), adds a resale view with hours_to_game. Grain: one row per game per hour.
SCHEMA · GRAIN
ETL

6

Analysis
SQL for pacing and curves; pandas + statsmodels for the regression; matplotlib for memo charts.
CTES · WINDOWS
REGRESSION

7

Outputs
Tableau Public dashboard, 1-page memo, README, LinkedIn post, outreach emails.
TABLEAU
MEMO

DESIGN DECISIONS YOU SHOULD BE ABLE TO DEFEND

Decision	Why
Hourly, not daily	Postseason games are on sale for 2–7 days. A daily snapshot gives 3 points per game; hourly gives 70+. The price curve only exists at this resolution.
All 12 teams, not just the Yankees	If the Yankees lose in three, a NY-only project is dead on Oct 1. League-wide, elimination is a <i>finding</i> , not a failure. Also gives ~45 games instead of 3.
CSV in git, not a database on a server	Free, auditable, reproducible by anyone who clones the repo. Recruiters can see the data grew hour by hour. SQLite is rebuilt from the CSVs on demand.
Two sources, not one	Resale alone shows demand. Primary alone shows the club's guess. The gap between them is the business question.
Giants/Jets in the background	Same scraper, different query, zero marginal cost. Becomes the December extension (a 12-week NFL price curve with a same-stadium control).
Regex on titles for "postseason"	Neither API has a clean postseason flag. Titles do ("ALDS", "Home Game 2"). Pragmatic, documented in code, and a good interview story about messy real data.

KNOWN LIMITS (SAY THESE BEFORE ANYONE ASKS)

- **Listings, not sales.** SeatGeek stats describe what's asked, not what cleared. Listing count falling is the best available proxy for sales.
- **Aggregates, not sections.** Lowest/average/highest across the whole building. Section-level data isn't public. Say so, then note what the club could do with its own Archtics data.
- **Fees.** Ticketmaster priceRanges may or may not include fees (the price_type field says). Compare like with like.
- **Scheduled runs can drift.** GitHub may delay a cron by minutes. We store the true captured_at and compute hours-to-game from it, so drift is harmless.
- **Time-TBD games.** Before a series is set, games carry placeholder times. The time_tbd flag lets us exclude or study them.

THE REPO A HIRING MANAGER WILL OPEN

```
Pricing-October/  
├── README.md           question · data · method · fir  
├── scrape.py           hourly collector (both APIs)  
├── build_db.py         CSV → SQLite, adds resale view  
├── analysis/  
│   ├── 01_pacing.sql   series pacing, hours-to-game o  
│   ├── 02_gap.sql      primary vs resale gap by game  
│   └── 03_series_state.sql price response to game outcom  
├── model.ipynb         regression + charts for the me  
├── dashboard/  
│   └── memo/pricing-october.pdf Tableau Public link + screensh  
├── data/*.csv          the append-only snapshot log  
└── .github/workflows/scrape.yml
```

2026 postseason bracket · fill it in as it happens

Seeds and records final as of Sep 27. Format: 1 and 2 seeds get byes; the 1 seed plays the 4/5 winner, the 2 seed plays the 3/6 winner. Best-of-3 · best-of-5 · best-of-7 · best-of-7. Pencil in the winner and the series score (e.g. “NYY 2–1”).

AMERICAN LEAGUE



SERIES TRACKER

Series	Winner	Score	Games	One thing the market did
AL WC · Red Sox at Yankees			__ of 3	
AL WC · White Sox at Astros			__ of 3	
NL WC · Phillies at Braves			__ of 3	
NL WC · Cubs at Padres			__ of 3	
ALDS · Rays vs			__ of 5	
ALDS · Guardians vs			__ of 5	
NLDS · Brewers vs			__ of 5	
NLDS · Dodgers vs			__ of 5	
ALCS			__ of 7	
NLCS			__ of 7	
World Series			__ of 7	

WHY THE BRACKET IS DATA

Every series outcome is an event in the dataset. When the Yankees win Game 1, the Game 3 “if necessary” listing either firms up or evaporates. When a series ends in a sweep, an entire game’s worth of listings goes to zero. The tracker column on the left is your annotation layer: what you noticed as a fan becomes the caption on the dashboard.

WATCH FOR THESE MOMENTS

- **Sweep.** Game 3 listings the morning after a 2–0. How fast do sellers cut?
- **Forced Game 5 / Game 7.** The scarcest ticket of the round. Price vs. Game 1 of the same series.
- **Opponent revealed.** The 1-seed’s Game 1 listing exists before the opponent is known. Does the price move when it is?
- **Yankees–Red Sox.** Rivalry premium: compare to the other three Wild Card hosts at the same hours-to-game.
- **Weather.** A postponement is a natural experiment on “time to game.”

Games actually played: ___ / 45 possible. That number belongs in the memo: it’s the size of the “if necessary” problem.

Pipeline health log · once a week, from the repo’s Actions tab: number of runs, failures, and the row counts printed in the log					
Week of	Runs	Failures	Rows (SeatGeek)	Rows (Ticketmaster)	Gaps noticed
Sep 28					
Oct 5					
Oct 12					
Oct 19					
Oct 26					

Schedule • Sep 29 → Oct 31 • every possible game

Wild Card and Division Series dates are official (MLB.com). LCS and World Series follow MLB’s published windows; confirm exact dates on mlb.com/postseason once set. Tick when played; strike through when not needed. Note column: anything you’d want on a chart. Home field in the WS goes to the better regular-season record.

WILD CARD • BEST OF 3 • ALL AT THE HIGHER SEED

	Date	Game	Note
☐	Tue Sep 29	Phillies at Braves G1 • 2p ET • Truist Park	
☐	Tue Sep 29	White Sox at Astros G1 • 5p ET • Daikin Park	
☐	Tue Sep 29	Red Sox at Yankees G1 • 8p ET • Yankee Stadium	
☐	Tue Sep 29	Cubs at Padres G1 • 10p ET • Petco Park	
☐	Wed Sep 30	Phillies at Braves G2 • 2p	
☐	Wed Sep 30	White Sox at Astros G2 • 5p	
☐	Wed Sep 30	Red Sox at Yankees G2 • 8p	
☐	Wed Sep 30	Cubs at Padres G2 • 10p	
☐	Thu Oct 1	Phillies at Braves G3 • <i>if necessary</i>	
☐	Thu Oct 1	White Sox at Astros G3 • <i>if necessary</i>	
☐	Thu Oct 1	Red Sox at Yankees G3 • <i>if necessary</i>	
☐	Thu Oct 1	Cubs at Padres G3 • <i>if necessary</i>	

DIVISION SERIES • BEST OF 5 • 2-2-1 • HIGHER SEED HOSTS G1, G2, G5

	Date	Game	Note
☐	Sat Oct 3	ALDS A & B G1 • at Rays, at Guardians	
☐	Sat Oct 3	NLDS A & B G1 • at Brewers, at Dodgers	
☐	Sun Oct 4	NLDS A & B G2	
☐	Mon Oct 5	ALDS A & B G2	
☐	Tue Oct 6	NLDS A & B G3 • at lower seed	
☐	Wed Oct 7	ALDS A & B G3 • at lower seed	
☐	Wed Oct 7	NLDS A & B G4 • <i>if necessary</i>	
☐	Thu Oct 8	ALDS A & B G4 • <i>if necessary</i>	
☐	Fri Oct 9	NLDS A & B G5 • <i>if necessary</i> • at higher seed	
☐	Sat Oct 10	ALDS A & B G5 • <i>if necessary</i> • at higher seed	

LCS • BEST OF 7 • 2-3-2 • WINDOW OCT 11–20

	Date	Game	Note
☐	___ Oct __	ALCS G1 • at higher seed	
☐	___ Oct __	NLCS G1 • at higher seed	
☐	___ Oct __	ALCS G2	
☐	___ Oct __	NLCS G2	
☐	___ Oct __	ALCS G3 • at lower seed	
☐	___ Oct __	NLCS G3 • at lower seed	
☐	___ Oct __	ALCS G4	
☐	___ Oct __	NLCS G4	
☐	___ Oct __	ALCS G5 • <i>if necessary</i>	
☐	___ Oct __	NLCS G5 • <i>if necessary</i>	
☐	___ Oct __	ALCS G6 • <i>if necessary</i> • at higher seed	
☐	___ Oct __	NLCS G6 • <i>if necessary</i> • at higher seed	
☐	___ Oct __	ALCS G7 • <i>if necessary</i>	
☐	___ Oct __	NLCS G7 • <i>if necessary</i>	

WORLD SERIES • BEST OF 7 • 2-3-2 • OCT 23–31

	Date	Game	Note
☐	Fri Oct 23	WS G1 • at _____	
☐	Sat Oct 24	WS G2	
☐	Mon Oct 26*	WS G3 • at _____	
☐	Tue Oct 27*	WS G4	
☐	Wed Oct 28*	WS G5 • <i>if necessary</i>	
☐	Fri Oct 30*	WS G6 • <i>if necessary</i>	
☐	Sat Oct 31	WS G7 • <i>if necessary</i>	

* Projected from MLB’s standard 2-3-2 pattern with off days after G2 and G5. G1 (Oct 23) and the G7 date (Oct 31) are published; confirm the middle dates when the WS schedule is released.

LOGGING A GAME IN RESULTS .CSV

One row per game, added the morning after. Cancelled “if necessary” games get a row too, with played = 0; those rows are the whole point.

series	game	date	home	away	played	hw	h1	winner	note
ALWC1	1	2026-09-29	NY Yankees	BOS Red Sox	1	0	0	NY Yankees	
ALWC1	2	2026-09-30	NY Yankees	BOS Red Sox	1	1	0	BOS Red Sox	rain 40m
ALWC1	3	2026-10-01	NY Yankees	BOS Red Sox	1	1	1	NY Yankees	
NLWC1	3	2026-10-01	ATL Braves	PHI Phillies	0	2	0		sweep

hw / h1 = home team’s series wins/losses *going into* the game (what the market knew). Use full team names in the real file so they join to SeatGeek’s home_team. note: anything a chart caption might want.

Data dictionary · what one row means

A hiring manager's first question about any dataset is “what's the grain?” Answer: **one row per game per hour per source**. Everything below hangs off that.

TABLE SEATGEEK · RESALE MARKET

Column	Type	Meaning
captured_at	TEXT (ISO)	UTC timestamp of the snapshot. Part of the key.
bucket	TEXT	mlb_postseason or nfl_metlife
event_id	TEXT	SeatGeek's stable id for the game. Part of the key.
title	TEXT	e.g. “ALDS: Yankees at Rays – Home Game 1 (If Necessary)”. Round and if-necessary are parsed from here.
datetime_local	TEXT	Scheduled first pitch, venue local time. Basis for hours_to_game.
time_tbd	BOOL	True while the start time is a placeholder.
home_team / away_team	TEXT	From the performers array (home_team flag). Away is null until the opponent is set.
listing_count	REAL	Number of active resale listings. Our demand/supply proxy.
lowest_price	REAL	Cheapest listing, USD. The “get-in” price.
average_price	REAL	Mean of listings.
median_price	REAL	Median of listings. Use this in the memo (robust to a \$5,000 dugout seat).
highest_price	REAL	Most expensive listing.

TABLE TICKETMASTER · PRIMARY MARKET

Column	Type	Meaning
captured_at	TEXT	Same clock as above; lets the two tables join on the hour.
event_id	TEXT	Ticketmaster's id. Does not match SeatGeek's . Join on (home_team, local_date) instead.
name	TEXT	Club's event name, e.g. “ALDS Home Game 2: Yankees v. TBD”.
local_date / local_time	TEXT	Scheduled date and time.
status	TEXT	onsale · offsale · cancelled · postponed · rescheduled. “Cancelled” is how an unneeded Game 5 dies.
price_type	TEXT	“standard” or “standard including fees”. Compare like with like.
min_price / max_price	REAL	The club's posted price range. min_price vs resale lowest_price is the gap.

DERIVED FIELDS (BUILT IN SQL, NOT STORED)

Field	Definition
hours_to_game	(julianday(datetime_local) – julianday(captured_at)) × 24. The x-axis of every price curve.
round	CASE on title: Wild Card / ALDS / NLDS / ALCS / NLCS / World Series.
if_necessary	title LIKE '%If Necessary%'. The core flag of the project.
series_state	Home team's series record at capture time, joined from a small hand-maintained results.csv you fill from the bracket page. Values like “1-0”, “0-2”, “tied 2-2”.
was_played	1 if the game happened; 0 if cancelled. From results.csv.
gap_pct	(resale_lowest – primary_min) / primary_min. Positive = market values it above the club's floor.
listing_delta_24h	Window function: listing_count minus its value 24 rows earlier for the same event. Sales velocity proxy.

THE ONE FILE YOU MAINTAIN BY HAND

data/results.csv · one row per game · columns: series, game_no, date, home, away, played (1/0), home_w_before, home_l_before, winner, note. Fill it from the bracket page after each game. It's tiny (≤45 rows), and it's what turns a price log into an event study. Recruiters like seeing a human-curated dimension table next to machine-collected facts; it shows you know the difference.

VOLUME ESTIMATE

Quantity	Estimate
Postseason games listed at any one time	10–30
Snapshots per day (both sources)	24 × 2
Rows per day	~600–1,500
Rows by Nov 1	~30,000–50,000
CSV size by Nov 1	~8–15 MB
Giants/Jets background rows	~10/hour, ~25,000 by January

Small enough for SQLite and Tableau Public with no tricks. Big enough that you can't eyeball it, which is the point.

INTERVIEW LINE

“The grain is game-hour. Facts come from two APIs on an hourly cron; the one dimension I couldn't get from an API, series state, I maintained by hand and joined in. That's the same shape as a team's warehouse: Archtics facts, a hand-kept event calendar.”

Analysis plan · seven questions, each with a method and a skill

Ordered from “works with three days of data” to “needs the whole postseason.” Q1–Q3 are done by Fall Break; Q4–Q6 need the LCS; Q7 is the memo’s headline. For a hiring manager: Q1 is the chart they’ll remember, Q3 the number they’ll quote, Q5 the one they’ll argue with, Q7 the proof you can model.

#	Question	Method	Output	Skill proved
1	The decay curve. How does the get-in price move from 72 hours out to first pitch?	Median lowest_price by hours_to_game bucket (72/48/24/12/6/3/1), per round. Plot every game as a faint line, the median as a bold one.	“Price curve” chart. First finding by Oct 2.	GROUP BY · CASE LINE CHART
2	The rivalry premium. Was Yankees–Red Sox priced above the other three Wild Card hosts at the same hours-to-game?	Same-hour comparison across the four WC series. Ratio of NYY median to the mean of the other three, by hour.	One number (“1.9× at 24h out”) plus a small multiple chart.	SELF-JOIN MATCHUP PRICING
3	The “if necessary” discount. How much cheaper is a Game 3/5/7 listing than a guaranteed game in the same series, before anyone knows it’ll happen?	Pair each if-necessary game with Game 1 of its series at matched hours-to-game. Compute discount %. Then track what happens once the game becomes certain or dies.	Discount table by round. The project’s signature stat.	CTES · MATCHED PAIRS UNCERTAINTY PRICING
4	Series-state elasticity. After a home win in Game 1, what happens to Game 2 listings overnight? After a loss?	Join results.csv. Compare median price and listing_count 2 hours before Game 1 vs 8 hours after, split by outcome. Window function for the “before/after” lag.	Before/after bar chart. Interview story about event studies.	LAG · WINDOWS EVENT STUDY
5	The primary-vs-resale gap. Where did the club price below the market, and by how much?	Join ticketmaster to seatgeek on (home_team, date, hour). gap_pct by game and round. Rank games by gap.	Ranked table + gap heat map (team × round).	CROSS-SOURCE JOIN DYNAMIC PRICING
6	Sell-through velocity. Does the listing count fall faster for some games, and does price rise when it does?	listing_delta_24h via window function; correlate with next-hour price change. Simple, honest about causality.	Scatter + correlation. A “velocity” view on the dashboard.	WINDOW FUNCTIONS CORRELATION
7	What explains price? One model, all games.	OLS on log(median_price): hours_to_game (log), round, if_necessary, home_seed, series_state, rivalry flag, weekend, home market size (MSA population). statsmodels, robust SEs, clustered by game.	Coefficient table with plain-English translation. The memo’s “so what.”	PANDAS · STATSMODELS REGRESSION · INFERENCE

FINDINGS TEMPLATE · WRITE THE SENTENCE BEFORE YOU HAVE THE NUMBER

One sentence, one number, one comparison. Draft with blanks on Oct 9, fill on Nov 1. If the data disagrees with you, that’s a better finding, not a failure.

Q	Expected sentence	If the data disagrees, the sentence becomes...
1	“Get-in prices fell ___% from 48 hours out to first pitch, and ___% of that drop came in the final 6 hours.”	“Prices <i>rose</i> into first pitch for ___ of ___ games: sellers held, buyers blinked. Clubs pricing to the 48-hour market underpriced.”
2	“Yankees–Red Sox Game 1 traded at ___× the median of the other three Wild Card hosts at the same hour.”	“The rivalry premium was smaller than the seed premium: ___ at ___× beat NYY. Matchup matters less than home record.”
3	“‘If necessary’ listings were ___% cheaper than Game 1 of the same series until the game was confirmed, then re-rated within ___ hours.”	“The market priced if-necessary games at near parity (___% discount): sellers already assumed a long series. Clubs can price them as certain.”
4	“A home win in Game 1 lifted Game 2 median prices ___% overnight; a loss cut them ___%.”	“Game 1 results barely moved Game 2 prices (±___%): demand is for the event, not the standings. Series-state pricing isn’t worth the complexity.”
5	“At 24 hours out, the resale floor sat ___% above the club floor for ___ of ___ games, largest for ___.”	“Resale floors sat <i>below</i> primary for ___ games: clubs overpriced late inventory and the secondary market cleared it cheaper.”
6	“Games that lost ___% of listings in 24 hours saw prices rise ___% the following hour.”	“Listing counts fell without prices rising: sellers de-listed rather than sold. Listing count is a weak sales proxy here; say so.”
7	“Holding round and time constant, if-necessary cost ___%, rivalry added ___%, and each series win added ___%.”	Report near-zero coefficients honestly, with the interval. “Can’t distinguish from zero with one postseason” is a fine sentence.

The model · one regression, explained in dollars

Q7 from the analysis plan: the part that separates “I made a dashboard” from “I can tell you what drives price.” One model, every choice defended, every coefficient translated into a ticket.

THE REGRESSION, SKETCHED

```
import pandas as pd, numpy as np, sqlite3
import statsmodels.formula.api as smf

con = sqlite3.connect("data/pricing.db")
df = pd.read_sql("""
    SELECT r.*, s.home_w_before AS w,
           s.home_l_before AS l, s.played
    FROM resale r
    LEFT JOIN results s
    ON s.home = r.home_team AND s.date = date(r.datetime_local)
    WHERE r.bucket = 'mlb_postseason'
    AND r.hours_to_game BETWEEN 1 AND 168
""", con)

df["log_p"] = np.log(df.median_price)
df["log_h"] = np.log(df.hours_to_game)
df["ifnec"] = df.title.str.contains("If Necessary").astype(int)
rounds = r"(Wild Card|ALDS|NLDS|ALCS|NLCS|World Series)"
df["round"] = df.title.str.extract(rounds)
nyy = df.home_team.eq("New York Yankees")
bos = df.away_team.eq("Boston Red Sox")
df["rival"] = (nyy & bos).astype(int)

m = smf.ols("log_p ~ log_h + C(round) + ifnec + rival + w + l",
            data=df).fit(cov_type="cluster",
                        cov_kws={"groups": df.event_id})
print(m.summary())
# ifnec = -0.35 → "if necessary" listings ~30% cheaper, all else equal
```

HOW TO READ A COEFFICIENT OUT LOUD

On a log-price model, a coefficient of -0.35 on `ifnec` means an “if necessary” listing is about 30% cheaper than a guaranteed game in the same round at the same hours-to-game ($e^{-0.35} \approx 0.70$). Say it that way in the memo. Never say “statistically significant” without saying what the number means for a ticket.

WHAT MAKES THIS MORE THAN A CLASS PROJECT

- **Clustered standard errors by game.** 70 hourly rows of one game aren’t 70 independent observations. Knowing that is the difference between a BA student and an analyst.
- **Log-log on hours.** Turns “price falls as the game nears” into an elasticity a pricing manager can use.
- **A hand-built dimension (series state).** Shows you understand data modeling, not just querying.
- **Limits stated first.** Listings ≠ sales; building-level aggregates; one postseason. Then the recommendation anyway.

VARIABLES, AND WHY EACH ONE IS IN

Variable	Definition	Why it belongs · what you expect
<code>log_h</code>	log(hours to first pitch)	Time-to-event is the dominant driver of resale prices. Log form makes the coefficient an elasticity: “a 10% cut in time left moves price X%.” Expect negative.
<code>C(round)</code>	Wild Card / DS / LCS / WS	Stakes rise by round. Fixed effects absorb that so the other coefficients aren’t just “the World Series is expensive.” Expect monotonic increase.
<code>ifnec</code>	1 if the title says “If Necessary”	The project’s headline variable: the uncertainty discount. Expect -0.2 to -0.5 (20–40% cheaper).
<code>rival</code>	1 for Yankees–Red Sox	The rivalry premium, controlled for round and time. Expect positive; the size is the finding.
<code>w, l</code>	Home team’s series wins / losses before this game	Series state. A team down 0–2 hosting Game 3 is a different product than one up 2–0. Expect <code>w</code> positive, <code>l</code> negative, and larger in magnitude late in the series.

CHECKS YOU RUN BEFORE YOU BELIEVE IT

- Plot residuals vs `hours_to_game`: if there’s a curve, add `log_h2`.
- Drop the World Series and re-fit: do the signs hold?
- Swap median for lowest price: same story?
- Cluster by game (done) and by series: do the standard errors change much?

WHAT YOU SAY IF SOMEONE ASKS “WHY NOT MACHINE LEARNING?”

“The goal was a number a pricing manager can act on, not the best predictor. Log-linear gives an interpretable elasticity and a clean uncertainty discount. For a forecast I’d try the same features in a gradient-boosted model and compare out-of-sample error.” That answer gets you hired more often than a random forest does.

The SQL you will write · in the order you're learning it

Your game plan has you at basic SELECT/JOIN now, CTEs by Oct 21, window functions by Oct 28. This page lines the project's queries up against that path, so every lesson lands on real data the same week.

WEEK OF SEP 28 · SELECT · WHERE · GROUP BY

```
-- Q0: is the pipeline healthy? rows per hour, last 24 h
SELECT substr(captured_at, 1, 13) AS hour,
       COUNT(*) AS games,
       SUM(lowest_price IS NULL) AS missing_price
FROM   seatgeek
WHERE  bucket = 'mlb_postseason'
      AND captured_at >= datetime('now', '-1 day')
GROUP BY 1 ORDER BY 1;
```

```
-- Q1: get-in price by hours-to-game bucket, by round
SELECT CASE WHEN title LIKE '%Wild Card%' THEN 'WC'
           WHEN title LIKE '%LDS%' THEN 'DS'
           WHEN title LIKE '%LCS%' THEN 'LCS'
           WHEN title LIKE '%World Series%' THEN 'WS'
           END AS round,
       CASE WHEN hours_to_game > 48 THEN '48h+'
           WHEN hours_to_game > 24 THEN '24-48h'
           WHEN hours_to_game > 6 THEN '6-24h'
           ELSE '0-6h' END AS bucket,
       COUNT(*) AS n, ROUND(AVG(lowest_price)) AS avg_get_in
FROM   resale
GROUP BY 1, 2 ORDER BY 1, 2;
```

FALL BREAK OCT 8-9 · JOIN · SUBQUERIES

```
-- Q2: Yankees-Red Sox vs the other three WC hosts, same hour
SELECT a.captured_at,
       a.lowest_price AS nyy,
       ROUND(AVG(b.lowest_price)) AS others,
       ROUND(a.lowest_price / AVG(b.lowest_price), 2) AS ratio
FROM   resale a
JOIN   resale b
      ON b.captured_at = a.captured_at
      AND b.title LIKE '%Wild Card%'
      AND b.home_team <> a.home_team
WHERE  a.home_team = 'New York Yankees'
      AND a.title LIKE '%Wild Card%Game 1%'
GROUP BY a.captured_at ORDER BY 1;
```

```
-- Q5: primary vs resale gap, per game, latest snapshot
SELECT s.home_team, date(s.datetime_local) AS game_date,
       t.min_price AS club_floor,
       s.lowest_price AS market_floor,
       ROUND(100.0 * (s.lowest_price - t.min_price) / t.min_price, 1) AS gap_pct
FROM   resale s
JOIN   ticketmaster t
      ON t.local_date = date(s.datetime_local)
      AND t.name LIKE '%' || substr(s.home_team, instr(s.home_team, ' ')+1) || '%'
      AND substr(t.captured_at, 1, 13) = substr(s.captured_at, 1, 13)
WHERE  s.captured_at = (SELECT MAX(captured_at) FROM resale)
ORDER BY gap_pct DESC;
```

WEEK OF OCT 19 · CTEs

```
-- Q3: "if necessary" discount vs Game 1, matched on hours
WITH g AS (
  SELECT event_id, home_team, title,
         title LIKE '%If Necessary%' AS ifnec,
         CAST(hours_to_game AS INT) / 6 * 6 AS h6, -- 6h bins
         median_price,
         resale WHERE hours_to_game BETWEEN 1 AND 72
),
g1 AS (
  SELECT home_team, h6, AVG(median_price) AS p_game1
  FROM   g WHERE title LIKE '%Game 1%' GROUP BY 1, 2
)
SELECT g.home_team, g.h6,
       ROUND(AVG(g.median_price)) AS p_ifnec,
       ROUND(g1.p_game1) AS p_game1,
       ROUND(100.0 * (AVG(g.median_price) - g1.p_game1) / g1.p_game1, 1) AS discount_pct
FROM   g JOIN g1 USING (home_team, h6)
WHERE  g.ifnec = 1
GROUP BY 1, 2 ORDER BY 1, 2;
```

WEEK OF OCT 26 · WINDOW FUNCTIONS

```
-- Q6: sell-through velocity: listings vs 24h earlier
WITH w AS (
  SELECT event_id, home_team, captured_at, hours_to_game,
         listing_count, median_price,
         LAG(listing_count, 24) OVER (PARTITION BY event_id
                                     ORDER BY captured_at) AS lc_24h_ago,
         LAG(median_price, 1) OVER (PARTITION BY event_id
                                    ORDER BY captured_at) AS p_prev
  FROM   resale
)
SELECT home_team, captured_at,
       listing_count - lc_24h_ago AS listings_delta_24h,
       ROUND(100.0 * (median_price - p_prev) / p_prev, 2) AS price_chg_pct,
       RANK() OVER (PARTITION BY substr(captured_at, 1, 10)
                   ORDER BY listing_count - lc_24h_ago) AS fastest_seller_today
FROM   w
WHERE  lc_24h_ago IS NOT NULL
ORDER BY captured_at DESC, fastest_seller_today;
```

```
-- Q4: Game 2 price before/after the Game 1 result
WITH r AS (
  SELECT s.*, x.winner = x.home AS home_won
  FROM   resale s
  JOIN   results x
      ON x.home = s.home_team AND x.game_no = 1
      AND s.title LIKE '%Game 2%'
), t AS (
  SELECT home_team, home_won,
         AVG(CASE WHEN hours_to_game BETWEEN 26 AND 28
                  THEN median_price END) AS before_g1,
         AVG(CASE WHEN hours_to_game BETWEEN 14 AND 16
                  THEN median_price END) AS after_g1
  FROM   r GROUP BY 1, 2
)
SELECT home_won, COUNT(*) AS series,
       ROUND(AVG(100.0 * (after_g1 - before_g1) / before_g1), 1) AS avg_move_pct
FROM   t GROUP BY 1;
```

Save each under analysis/ with the question and, later, the answer as a comment. That folder is what you walk a hiring manager through.

Dashboard spec · Tableau Public · four views, one story

A dashboard is an argument, not a gallery. Four views, top-left to bottom-right, each answering the question the previous one raises. One data source: `resaLe` joined to `results`, exported from SQLite as a single CSV.

VIEW 1 · TOP LEFT

The decay curve

Line chart. X = hours to first pitch (reversed, 72 → 0). Y = median get-in price. One faint line per game, bold median per round, colored by round. Annotations: “Yankees–Red Sox G1”, “Game 3 cancelled”.

Question it answers: when should a fan buy, and when should a club reprice?

line chart · hours-to-game × price

VIEW 2 · TOP RIGHT

The “if necessary” discount

Bar chart by round: median discount of if-necessary listings vs Game 1 of the same series, at matched hours-to-game. Second bar: what those listings did once the game was confirmed (or cancelled).

Question it answers: how much does uncertainty cost, and who bears it?

grouped bars · round × discount %

VIEW 3 · BOTTOM LEFT

The gap map

Heat map: teams (rows) × rounds (columns), cell = primary-vs-resale gap % at 24 hours out. Red = market above club floor by a lot. Tooltip shows club floor, market floor, listing count.

Question it answers: where did clubs leave money on the table?

heat map · team × round

VIEW 4 · BOTTOM RIGHT

Series state

Before/after dumbbell chart: each Game 2, price 27 h before Game 1 vs 15 h after, colored by whether the home team won Game 1. Small, honest, memorable.

Question it answers: does last night’s score move today’s price?

dumbbell · before vs after, by outcome

RULES FOR THE BUILD

- **One headline per view**, written as a sentence in the title: “Get-in prices fall 40% in the last 24 hours,” not “Price vs Time.”
- **Filters**: round, team, if-necessary. Nothing else. Three filters is a dashboard; ten is a spreadsheet.
- **Median, not mean**, for price. State it in a footnote.
- **Color = round** everywhere, same palette in all four views and in the memo charts.
- **A “how to read this” caption** under each view, one line, for the person who’s never seen a dumbbell chart.
- **Source line**: “SeatGeek Platform API and Ticketmaster Discovery API, hourly snapshots Sep 28 – Nov 1, 2026. Listings, not sales.”

BUILD ORDER

When	Do
Fri Oct 16	Tableau Public account. Export <code>resaLe</code> to CSV. Connect. View 1 (the decay curve). Publish it even if it’s ugly.
Sat Oct 17	View 2 and View 3. Dashboard layout, filters, titles as sentences.
Sat Oct 24	View 4 once the LCS has given you enough Game 2s. Captions, source line, final palette.
Wed Nov 18	Rebuild the same four views in Power BI (already in your plan). Same story, second tool, one more line on the resume.

SCREENSHOT CHECKLIST FOR THE README

- Full dashboard, 1600 px wide, light background
- Each view alone, cropped, with its sentence-title visible
- One GIF of the round filter changing (optional, 5 seconds)

Pricing October · Blueprint v1

Page 10 · Dashboard spec

The memo and the outreach · where the project earns its keep

Analysts who get hired write one page that a VP reads in four minutes and forwards. Here is that page's skeleton, and the three messages that carry it to the people you want to work for.

MEMO · ONE PAGE · “PRICING OCTOBER”

To: VP, Ticketing Strategy & Analytics · **From:** Michael Schwartz · **Date:** Nov 2, 2026

Re: What the resale market told us about postseason pricing, and three things to do about it

Bottom line (3 lines). The finding, the number, the recommendation. Written last, placed first.

Question (2 lines). How does the market price games that may not happen, and where is our floor below theirs?

Data (3 lines). Hourly snapshots, two sources, dates, games, rows. Listings not sales. Building-level aggregates.

Method (3 lines). Matched comparisons at equal hours-to-game; one regression with clustered errors. Plain words.

Findings (3 bullets, one chart each). The decay curve. The if-necessary discount. The gap map.

Limits (3 bullets). Say them before they're asked.

Recommendations (3 bullets, each with a number). e.g. “Hold if-necessary inventory off primary sale until the series state is known; the market discounts it 30% and re-rates it within 6 hours of certainty.”

Appendix link. Dashboard, repo, regression table.

MEMO RULES

- Every number has a comparison next to it. “\$212” means nothing; “\$212, 1.9× the other Wild Card hosts at the same hour” does.
- No jargon a season-ticket rep wouldn't use. “Elasticity” is fine if the next clause says what it means.
- Charts are the ones from the dashboard, cropped, with the sentence-titles. No new charts in the memo.
- Recommendations are things a club can actually do next October. Not “collect more data.”

MESSAGE 1 · PROJECT-LED OUTREACH (TEMPLATE H, ADAPTED) · NOV 3–5

Subject: A student's look at postseason ticket pricing (2-min read)

Hi [Name], I'm a junior in Business Analytics at Auburn. As a learning exercise I tracked resale and primary prices for every 2026 postseason game, hourly, from the Wild Card round through the World Series: [dashboard link]. One finding: “if necessary” listings traded about [X]% below guaranteed games in the same series until the night before, then re-rated within hours. Your internal data surely tells a fuller story. I'd value your critique of the approach, even one line. Thank you, Michael · [LinkedIn] · [repo]

Send to: Mets Ticketing Strategy & Analytics (Manahan's group, analysts first), Yankees Business Strategy, MLB League Office Analytics & Strategy, Braves Business Strategy & Analytics, Rays Strategy & Analytics. Then any team whose games are in the dashboard. Stagger: no two people at the same club on the same day.

MESSAGE 2 · LINKEDIN POST · NOV 2

I spent October collecting a dataset instead of just watching baseball. Every hour, for every postseason game, I logged resale and primary ticket prices (two public APIs, a Python script, GitHub Actions doing the work while I was in class). ~40,000 rows later, three things stood out: [finding 1 with number] · [finding 2] · [finding 3]. Dashboard and code in the comments. Built with SQL, Python and Tableau; happy to hear where the method is wrong. #SportsBusiness #Analytics @Auburn Sports Analytics Club

MESSAGE 3 · APPLICATION ADD-ON (TEMPLATE F) · DEC

...I recently built an hourly pricing dataset for the 2026 MLB postseason (two APIs, SQL, a regression, Tableau) and wrote it up as a one-page memo on if-necessary inventory. It lines up with the posting's ticketing analytics work: [link].

The ask that matters most: after any reply, “Who else should I talk to?” Every October analyst knows three others. The project is the door; that question is the hallway.

Skills matrix · what a recruiter wants to see, and where this project shows it

Built from the actual postings on your target list (Colts, Stars, Nationals, Rays, Charlotte FC, Texans, Jaguars, Heat) and the NBA/NFL/MLB league programs. Left column is their language. Right column is your proof.

What postings ask for	Where the project proves it	The artifact you point to	Resume line
SQL (required in nearly every posting)	Six saved queries from GROUP BY to window functions on a 40k-row table you built. Cross-source join on messy keys.	analysis/*.sql	"SQL (SQLite): CTEs, window functions"
Python / R	Collector (requests, JSON flattening, error handling), loader (sqlite3), model (pandas, statsmodels).	scrape.py · build_db.py · model.ipynb	"Python: pandas, statsmodels, requests"
Tableau / Power BI	Four-view public dashboard with sentence titles; rebuilt in Power BI Nov 18.	Tableau Public link · Power BI screenshots	"Tableau Public, Power BI"
Ticketing & pricing analytics	Primary vs resale gap, decay curve, if-necessary inventory, series-state repricing. The vocabulary of a ticketing strategy group.	Memo findings 1–3	"Ticket pricing analysis: primary vs secondary"
CRM / data warehouse concepts	Fact tables from APIs + a hand-kept dimension (results.csv), explicit grain, indexes, a view layer. Same shape as Archtics → warehouse.	Data dictionary in README	"Designed fact/dimension schema"
Data collection / ETL	Hourly unattended pipeline for five weeks; append-only log; rebuildable database.	Actions run history (hundreds of green checks)	"Built hourly ETL pipeline (GitHub Actions)"
Statistics / modeling	Matched comparisons, event study, OLS with clustered SEs, coefficients translated to dollars.	Regression table in memo appendix	"Regression modeling of ticket prices"
Reporting to non-technical stakeholders	One-page memo to a VP; dashboard captions; LinkedIn post that a fan understands.	memo/pricing-october.pdf	"Executive memo and dashboard"
Attention to detail	Fees flag handled; time-TBD flag handled; cron drift handled; listings-vs-sales stated up front.	Limits section, README	(shows in interview)
Self-direction / "projects taken start to finish"	Question → pipeline → analysis → memo → outreach, alone, on a deadline set by the World Series.	The repo's commit history	"Independent project, Sep–Nov 2026"
Passion for the industry (without talking like a fan)	You spent October measuring the market instead of arguing about the bullpen. Talk revenue, inventory, fans-as-customers.	The LinkedIn post's first line	—
Computer science fundamentals (your minor)	Git workflow, CI/CD, secrets management, idempotent scripts, regex, pagination, rate-limit awareness, reproducible builds.	.github/workflows/scrape.yml · commit log	"CI/CD, git, API integration"

INTERVIEW QUESTIONS THIS PROJECT ANSWERS · AND THE SHAPE OF YOUR ANSWER

They ask	You say (sketch)
"Tell me about a data project you took start to finish."	Question → pipeline → analysis → memo → outreach. The 30-second version (p. 13), then stop and let them pick a thread.
"Walk me through your data model."	Grain is game-hour. Two fact tables from APIs, one hand-kept dimension (series state), a view with hours-to-game. Same shape as Archtics → warehouse.
"What was the hardest part?"	Joining two sources with no shared key (team + date + hour), and deciding what a cancelled Game 5 <i>means</i> in the data. Both are real-world data problems, not textbook ones.
"How do you know your result is real?"	Matched comparisons at equal hours-to-game; clustered errors by game; re-fit without the World Series; median vs lowest. And the limits: listings, not sales.
"Write a query that..." (live SQL)	You've written six on this schema. Pacing, gap, before/after, running totals: rehearse them from memory in December (p. 15).
"Why sports business analytics?"	Not "I love sports." "Fans are customers with strong preferences and the inventory is perishable to the hour. It's the hardest pricing problem I've seen; I spent October measuring it."

What it becomes on paper · resume, TeamWork Online, and the pitch

The project is only worth what you can say about it in thirty seconds, three bullets, and one headline. Write these on Nov 2 with the real numbers filled in; the placeholders here are the shape.

RESUME: THE PROJECTS ENTRY (ADD NOV 2)

Pricing October · Independent project · Sep – Nov 2026

- Built an hourly ETL pipeline (Python, GitHub Actions) collecting resale and primary ticket prices for all 2026 MLB postseason games from two public APIs; ~40k rows into SQLite
- Wrote SQL (CTEs, window functions) and a clustered-SE regression to measure the “if necessary” discount and the primary-vs-resale gap by round
- Published a 4-view Tableau Public dashboard and a one-page memo with three pricing recommendations; shared with analytics staff at 8 clubs

TEAMWORK ONLINE: THE EXPERIENCE ENTRY (ADD NOV 2)

Title: Sports Analytics Project Lead (Independent) · **Org:** Pricing October · **Dates:** Sep 2026 – Nov 2026

Same three bullets. Skills: Data Analysis, SQL, Python, Tableau, Ticketing, Data Engineering, Research, Project Management.

Headline update: “Business Analytics | Ticketing & Pricing Analytics | SQL · Python · Tableau | Auburn ’28”. SQL moves from “learning” to listed once Q3 is written.

LINKEDIN: THE PROJECTS SECTION

Pricing October: how the MLB postseason ticket market prices uncertainty · Sep 2026 – Nov 2026

Same three bullets, plus the dashboard link as the project URL and two media items: the full dashboard screenshot and the memo PDF. Skills: SQL, Python, Tableau, Data Engineering, Statistics.

HEADLINES · PICK ONE PER PLATFORM

- **TeamWork Online:** Business Analytics | Ticketing & Pricing Analytics | SQL · Python · Tableau | Auburn ’28
- **LinkedIn:** Business Analytics @ Auburn · built an hourly MLB postseason ticket-pricing dataset · SQL, Python, Tableau · Summer 2027 sports strategy & analytics
- **Email signature:** Michael Schwartz · Auburn Business Analytics ’28 · Pricing October: [dashboard link]

INTERVIEW: THE THREE-MINUTE WALK-THROUGH

1. **Question** (20 s): postseason inventory is high-margin and time-starved; how does the market price games that might not happen?
2. **Data** (40 s): two APIs, hourly, five weeks, unattended on GitHub Actions; ~40k rows; a hand-kept results table for series state. Listings, not sales.
3. **Method** (40 s): matched comparisons at equal hours-to-game; one log-price regression with game-clustered errors.
4. **Findings** (50 s): the three numbers, each with its comparison.
5. **What I’d do with club data** (30 s): section-level, actual sales, pace vs. last postseason, a hold-back rule for if-necessary inventory.

AFTER NOV 5

- **Nov 18:** Power BI rebuild (already in plan).
- **Dec 16–23:** Extension: Giants/Jets 12-week curve with same-stadium control. Same repo, second memo.
- **Dec 30:** Mock interview: “walk me through Pricing October” in 3 minutes, then in 30 seconds.
- **Spring:** If Auburn Athletics shares ticket data, the same pipeline shape and queries apply. That’s the pitch for the free dashboard project.

THE THIRTY-SECOND VERSION

“During the 2026 postseason I built a pipeline that logged resale and primary ticket prices every hour for every game, about forty thousand rows. I used SQL and a regression to measure how the market discounts games that might not be played and where clubs priced under the market, then wrote it up as a one-page memo with three recommendations and sent it to analytics staff at eight clubs.”

EXPLAINING IT TO A TICKET SALES MANAGER

“You know how a Game 5 that might not happen sells for less than Game 1? I measured how much less, for every series, every hour, and how fast it catches up once the game is confirmed. The short version: the market discounts uncertainty about __%, and it fixes that in about __ hours. So if the club holds that inventory until the series state is known, it sells at the higher number.” No SQL, no regression, one action. That’s the test of whether you understand it.

Day by day · Sep 27 → Nov 5 · project tasks only

This replaces the “Project 1” rows in your game plan. Networking, applications and SQL lessons stay where they are. The daily minimum still runs underneath. ⌚ = time estimate.

	Day	Task
☐	Sun Sep 27	Read this blueprint. Create SeatGeek + Ticketmaster keys (p. 2, items 1–2). ⌚ 15 min
☐	Mon Sep 28	Fill .env, create GitHub repo + secrets, say “keys are in.” First snapshot runs; fixes; push. First green check tonight. ⌚ 20 min
☐	Tue Sep 29	FIRST PITCH 2 pm ET. Check Actions at 1 pm and 9 pm. Watch how Yankees–Red Sox G1 prices move in the last 6 hours (View 1 preview). ⌚ 5 min
☐	Wed Sep 30	Actions check. Start results.csv: four rows for last night’s games. ⌚ 10 min
☐	Thu Oct 1	Game 3s: note which were played and which vanished; that’s your first “if necessary” observation. Fill bracket. ⌚ 10 min
☐	Fri Oct 2	Run build_db.py. Query Q0 (health) and Q1 (decay curve) on Wild Card data. Screenshot the first result; it goes in the README as “day 4.” ⌚ 40 min
☐	Sat Oct 3	Division Series G1. Actions check. Note whether the 1-seeds’ listings existed before the opponent was known (Q3 material). ⌚ 5 min
☐	Sun Oct 4	Weekly review. Pipeline health row (p. 5). results.csv up to date. ⌚ 15 min
☐	Mon–Wed Oct 5–7	Actions check daily. results.csv after each game. ⌚ 5 min/day
☐	Thu Oct 8 · Fall Break	3-hour block: Q2 (rivalry) and Q5 (gap). Save both under analysis/ with the answer in a comment. Write the data dictionary into the README. ⌚ 3 h
☐	Fri Oct 9 · Fall Break	2-hour block: Q3 (if-necessary discount) on DS data. Draft the three findings as sentences, even if the numbers will change. ⌚ 2 h
☐	Sat Oct 10	ALDS Game 5s, if any. Bracket. ⌚ 5 min
☐	Sun Oct 11	Weekly review. LCS begins. Pipeline health row. ⌚ 15 min
☐	Mon–Thu Oct 12–15	Actions check daily. results.csv. Oct 14: while drafting NBA answers, use this project for “tell me about a data project.” ⌚ 5 min/day

	Day	Task
☐	Fri Oct 16	Tableau Public account. Export resale CSV. View 1 built and published. ⌚ 1.5 h
☐	Sat Oct 17	Views 2 and 3. Dashboard layout. Sentence titles. ⌚ 2.5 h
☐	Sun Oct 18	Weekly review. Pipeline health row. ⌚ 15 min
☐	Mon–Fri Oct 19–23	Actions check daily. results.csv. Oct 21–22: Q4 and Q6 as your window-function homework (they’re the Kaggle lessons, on your data). WS G1 Fri Oct 23. ⌚ 5 min/day + 1 h
☐	Sat Oct 24	Memo draft (p. 11) with placeholder numbers. View 4. Send the draft to me for the VP read. ⌚ 2 h
☐	Sun Oct 25	Weekly review. Regression first pass (Q7) on everything through the LCS. ⌚ 1.5 h
☐	Mon–Thu Oct 26–29	WS games. Actions check. results.csv. Revise memo with real numbers as the WS adds rows. ⌚ 10 min/day
☐	Fri Oct 30	README: question, data, method, three findings with screenshots, limits, how to run. ⌚ 1.5 h
☐	Sat Oct 31	WS G7 if necessary. Final polish. Ask one club officer or professor for a critique. ⌚ 1 h
☐	Sun Nov 1	Final snapshot. Re-run build_db, re-run Q1–Q7, lock the numbers. Final memo PDF. Checkpoint from the game plan: ~40 messages sent, 5+ calls. ⌚ 2 h
☐	Mon Nov 2	PUBLISH. LinkedIn post, dashboard link, repo link. Resume Projects entry. TeamWork Online Experience entry. ⌚ 45 min
☐	Tue Nov 3	Outreach wave: Mets, Yankees, MLB (Message 1). ⌚ 30 min
☐	Wed Nov 4	Apply to whatever’s live (game plan). Attach the memo link. ⌚ per posting
☐	Thu Nov 5	Outreach wave: Braves, Rays, Panthers, Titans, Hornets, Jaguars. Stop the MLB scraper (leave Giants/Jets running). ⌚ 30 min

TOTAL PROJECT TIME, SEP 27 → NOV 5

About **28 hours** over six weeks: ~6 h in the two Fall Break blocks, ~8 h on the dashboard and memo weekends, ~4 h on the README and publish, and ~10 h of five-minute daily checks and results.csv. The pipeline does the other 800 hours.

Risks, fallbacks, and how we know it's working

RISK REGISTER

Risk	Odds	What we do
SeatGeek no longer issues client IDs	Medium	Ticketmaster only. Lose the resale side; keep the primary price curve, status changes (cancelled Game 5s) and the club-side story. Project still ships. Alternative resale source: a small manual daily log of 6 marquee games.
SeatGeek returns events but empty stats	Low-med	Known to happen for some events. We log the row anyway (nulls are data: "no listings yet") and filter in the resale view.
Title regex misses a postseason game or catches a non-postseason one	Medium	Day 1 check: print the distinct titles, adjust the regex, re-run. Documented in the README as a real-data lesson.
GitHub cron silently skips hours	Low	GitHub disables schedules on repos with no activity for 60 days; ours commits hourly, so no. Drift of minutes is handled by storing captured_at. Health query Q0 catches gaps.
Rate limits	Low	~12 calls per hour total. Ticketmaster allows 5,000/day. SeatGeek's limit is undocumented; hourly is conservative.
Yankees swept, NY hook gone	~50%	By design, the dataset is league-wide. The NY hook becomes "I tracked the Yankees-Red Sox rivalry premium" plus the Mets/Yankees analysts still care about postseason pricing. Nothing changes in the plan.
Short series everywhere, small dataset	Low	Even all-minimum series give 33 games and ~25k rows. If-necessary cancellations are themselves the finding.
You fall behind on SQL	Medium	Queries on p. 9 are written; you run and modify them rather than write from blank. Understanding > authorship for the first pass. Rewrite from memory in December as interview prep.
Midterms collide with Oct 16-17 or Oct 24	Medium	Dashboard views slide a week; memo slides to Oct 31. Nov 2 publish holds; it has a two-day buffer built in.

DEFINITION OF DONE · CHECK EVERY BOX BEFORE NOV 2

- ☐ Pipeline ran for 30+ consecutive days; gaps listed in README
- ☐ results.csv complete for every game played or cancelled
- ☐ Six SQL files in analysis/, each with the answer in a comment
- ☐ Regression notebook runs top to bottom from a clean clone
- ☐ Tableau Public dashboard: 4 views, sentence titles, source line, 3 filters
- ☐ Memo: one page, three findings with numbers, three recommendations, limits stated
- ☐ README: question · data dictionary · method · findings with screenshots · limits · how to run
- ☐ One outside critique received and one change made because of it
- ☐ LinkedIn post published; resume and TeamWork Online updated
- ☐ You can explain every line of scrape.py and every query without notes

SIGNALS IT'S WORKING (BEFORE ANYONE HIRES YOU)

- A reply to Message 1 from anyone at a club. One is enough; it becomes a call.
- A comment on the LinkedIn post from someone in the industry you don't know.
- An interviewer asks "how did you handle the games that got cancelled?" and you have a real answer.
- You catch yourself explaining hours-to-game elasticity to a friend at a bar.

IF EVERYTHING GOES WRONG · THE MINIMUM VIABLE VERSION

Say midterms eat October, one API dies, and the Yankees go out in two. This still ships on Nov 2 and still beats most student portfolios:

- One source, daily.** Ticketmaster primary price range and status for every postseason game, once a day. Still a real pipeline, still unattended.
- Two queries.** Q1 (price by days-to-game) and the status log of every if-necessary game that was cancelled: how many, how long they were on sale, at what price.
- One chart, one page.** A single decay chart in Tableau Public and a half-page memo: "___ of ___ if-necessary games were sold and then cancelled. Here's what that costs fans and what a hold-back rule would do."